MedBase Virtual Trials Lab

Run a simulated type 2 diabetes drug trial in your browser, then see how far to trust the answer.

The idea

Real drug trials take years and huge sums, so students and early researchers rarely get to see how a study behaves before it runs. Many health AI tools also give confident answers with no way to check them.

MedBase lets anyone rehearse a trial for free, on virtual patients built from real health survey data. You ask in plain English, an AI copilot plans the experiment, and a simulator computes every drug-effect number. Beside the answer, the lab shows its own track record, including where it was wrong.

What we built

- **Conversational AI copilot:** explains ideas, follows up on "why?", and runs a real simulation when you ask about a drug's effect.
- **Virtual trials:** four real drugs and hypothetical molecules, simulated in the visitor's own browser.
- **AI Trial Designer:** set a goal, a neural network proposes a study, and the simulator checks it.
- **Instant neural sliders:** a small network gives answers in milliseconds, and you can race it against the full simulator.
- **Vector database in the browser:** PostgreSQL with pgvector finds the right notes by meaning, with no server or account.
- **Safety guard and security:** personal and urgent messages stop before any AI, code checks the AI's figures, and the AI proxy accepts only the lab's own prompts.
- **Public blind test:** predictions were frozen before real trial results were opened, and every miss stays visible.

How a question flows

Your question → safety guard → AI copilot and vector database → simulator → code checks the reply → answer with its track record.

Results

What was tested	Result
Blind test on 20 real results from trials it never saw	Mean error 0.29 HbA1c points. 10 of 20 within 0.30. Passed 2 of 4 checks, and did not clearly beat a simple rival.
Neural surrogate on 480 simulator runs it never trained on	R^2 0.949, mean HbA1c error 0.082 points
AI Trial Designer across 36 goals	46% fewer simulator runs, same verdict on 36 of 36
Finding the right note, on 60 held-out questions	98.3% in the top 3
Messages that must be refused, on 70 blind-written examples	0 reached the AI

Technology

Python, NumPy, SciPy, emcee (Bayesian MCMC), Pyodide, React, TypeScript, three.js, PostgreSQL with pgvector through PGlite. NVIDIA Nemotron models on NIM for language and embeddings. Hosted on Hugging Face Spaces, with a small proxy on Vercel.

What it is not

It is a simulator, not medical advice, and it says nothing about any one person. It has not been clinically validated, and its virtual patients come from a US survey, not an Indian cohort. General answers from the copilot come from the AI model and can be wrong. Nobody outside the project has used it yet; the next step is a pilot with a class, and adding Indian patient data.

Try it

Live: https://zade06079-medbase-virtual-trials-lab.static.hf.space
Code: https://github.com/shreynachaegari-glitch/medbase-virtual-trials-lab